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7.Q.1 (1/1 point)

Suppose we want to fit a generalized additive model (with a continuous response) for y against X_1 and X_2 . Suppose that we are using a cubic spline with four knots for each variable (so our model can be expressed as a linear regression after the right basis expansion).

Suppose that we fit our model by the following three steps:

- 1) First fit our cubic spline model for y against X_1 , obtaining the fit $\hat{f}_1(x)$ and residuals $r_i = y_i - \hat{f}_1(X_{i,1})$.
- 2) Then, fit a cubic spline model for r against X_2 to obtain $\hat{f}_2(x)$.
- 3) Finally construct fitted values $\hat{y}_i = \hat{f}_1(X_{i,1}) + \hat{f}_2(X_{i,2})$.

Will we get the same fitted values as we would if we fit the additive model for y against X_1 and X_2 jointly?

- ☐ yes, no matter what
- ☐ only if X_1 and X_2 are uncorrelated
- ☒ not necessarily, even if X_1 and X_2 are uncorrelated. ✓

EXPLANATION

If X_1 and X_2 are uncorrelated, and we are only fitting a linear regression (only linear terms for X_1 X_2) then this method would work. However, even if X_1 and X_2 are uncorrelated, the nonlinear basis functions might be.

Hide Answer

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